

Whistleblowing in Engineering

Internal and External Whistleblowing

Introduction

Whistleblowing means reporting unethical, illegal, or dangerous activity done by an employer, supervisor, or organization. In engineering, whistleblowing usually happens when public safety, health, environment, or honesty is at serious risk.

According to Fleddermann, whistleblowing is connected with both the **rights** and **responsibilities** of engineers. Engineers have a duty to protect public health and safety, and they also have the right to report wrongdoing and expect proper action.

1. Internal Whistleblowing

Internal whistleblowing happens when the employee reports the problem **inside the organization**.

This may include reporting to:

- immediate supervisor
- higher management
- company president
- board of directors
- ethics committee

Internal whistleblowing is usually better as the first step because it gives the organization a chance to correct the problem. It is also considered less serious than going outside the company.

Example:

An engineer tells higher management that safety testing is being skipped before product release.

2. External Whistleblowing

External whistleblowing happens when the employee reports the problem **outside the organization**.

This may include reporting to:

- newspapers/media
- law-enforcement authorities
- government agencies
- professional bodies
- public safety regulators

External whistleblowing is more serious because it can damage the company's image and may be seen as disloyalty. However, it may become necessary if internal reporting fails and public safety is still at risk.

Example:

An engineer reports to a government safety authority that a company is releasing unsafe products after ignoring internal complaints.

3. Anonymous and Acknowledged Whistleblowing

Whistleblowing can also be **anonymous** or **acknowledged**.

In **anonymous whistleblowing**, the person does not reveal their name. This may protect the employee, but the complaint may look weaker because no one accepts responsibility for it.

In **acknowledged whistleblowing**, the person gives their name and stands behind the accusation. This is stronger, but it may create personal risk for the whistleblower.

4. When Should Whistleblowing Be Done?

Whistleblowing should not be done for small issues, revenge, anger, or personal benefit. It should be done only when the matter is serious and affects public interest.

Fleddermann gives four conditions before whistleblowing should be attempted:

1. Need

There must be a clear and important harm that can be avoided. The issue should be serious enough to justify whistleblowing.

2. Proximity

The engineer must have first-hand knowledge of the problem. Hearsay is not enough. The engineer should also have enough expertise to understand the issue.

3. Capability

The whistleblower must have a reasonable chance of success. There should be some possibility that reporting will stop the harmful activity.

4. Last Resort

Whistleblowing should be used only after other proper internal steps have failed, or when no better person is available to report the matter.

5. When Is Whistleblowing a Moral Duty?

An engineer may be **morally allowed** to blow the whistle when the four conditions are fulfilled.

But an engineer becomes **morally obligated** to blow the whistle when there is great and immediate danger to people if the harmful activity continues. In such cases, protecting public safety becomes more important than loyalty to the employer.

6. Correct Motive for Whistleblowing

The motive must be protection of public interest. Whistleblowing is ethical when it is done to prevent harm, danger, fraud, or serious wrongdoing.

It is not ethical if it is done for revenge, jealousy, personal fame, money, or future benefit. Fleddermann clearly says the whistleblower should understand their motives before taking this step.

How to Apply in Case Studies

In a case study, first identify the wrongdoing. Then check whether the issue involves public safety, fraud, false testing, environmental damage, or illegal action.

After that, apply the four conditions:

Need: Is the harm serious?

Proximity: Does the engineer have direct knowledge?

Capability: Can reporting stop the harm?

Last resort: Have internal options failed?

Then decide whether the engineer should use internal or external whistleblowing. Usually, the engineer should try internal reporting first. External whistleblowing should be used only if the danger is serious and the organization refuses to act.

Case Study: Robert and Safety Testing

Robert is asked by the CEO to skip required torpedo safety testing and falsify documents. This is serious because the product may become dangerous and the paperwork would be dishonest.

Robert should first refuse to sign the false documents and report the issue internally to higher management or the ethics office. The four conditions are present: there is serious harm, Robert has direct knowledge, reporting may stop the danger, and if internal reporting fails, whistleblowing becomes the last option.

If the company still forces him to falsify the test results, Robert should go for external whistleblowing to the proper authority because public safety is more important than his job.

Memorization Checklist

Remember whistleblowing with:

N-P-C-L

N — Need: Serious harm must exist.

P — Proximity: First-hand knowledge is needed.

C — Capability: Reporting should have chance of success.

L — Last resort: Use it after other options fail.

5-Mark Exam Answer

Whistleblowing means reporting unethical, illegal, or dangerous activity by an employer or organization. According to Fleddermann, it is connected with the rights and responsibilities of engineers because engineers have a duty to protect public safety and a right to report wrongdoing. Internal whistleblowing means reporting the issue inside the organization, such as to higher management, the president, or the board of directors. External whistleblowing means reporting outside the organization, such as to media, law-enforcement, or government authorities.

Whistleblowing should be attempted only when four conditions are fulfilled: need, proximity, capability, and last resort. Need means there must be serious harm. Proximity means the engineer must have first-hand knowledge and enough expertise. Capability means reporting should have a reasonable chance of stopping the harm. Last resort means other internal options should be tried first. Therefore, engineers should usually try internal whistleblowing first, but if the organization refuses to act and public safety is in danger, external whistleblowing becomes ethically justified.